

Fig. 3: An actual-size, single-side PCB for the rotation counter

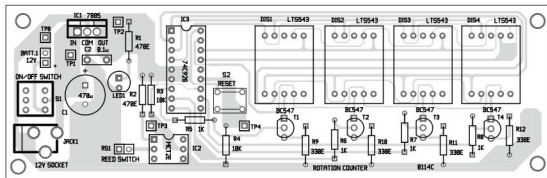


Fig. 4: Component layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- IC2 - MCT2E opto-coupler
- IC3 - 74C926 4-digit counter
- T1-T4 - BC547 npn transistor
- DIS1-DIS4 - LT5543 7-segment display
- LED1 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 470-ohm
- R3, R4 - 10-kilo-ohm
- R5-R8 - 1-kilo-ohm
- R9-R12 - 330-ohm

Capacitors:

- C1 - 470 μ F, 25V electrolytic
- C2 - 0.1 μ F ceramic disk

Miscellaneous:

- BATT.1 - 12V battery
- JACK1 - 12V, 500mA DC adaptor
- S1 - On/off switch
- S2 - Push-to-on switch
- RS1 - Reed switch
- Small magnet

DIS4), reed switch (connected at RS1) and a small magnet.

Opto-coupler IC2 is used to isolate the counter and provide a clock pulse to counter IC3. It counts a pulse when contacts of the reed switch touch each other due to the magnet that comes near it at each rotation. At each rotation, the magnet and reed switch

face each other once and the contacts get closed, which causes LED in the opto-coupler to turn on and drive the transistor. This brings clock pulse equal to the number of rotations at the CLK pin 12 of 4-digit counter IC3. IC3 is used to count the clocks received by opto-coupler IC2 and display them on the four common-cathode 7-segment displays (DIS1-DIS4).

IC3 has a 4-digit counter, an internal output latch, npn output sourcing drivers for a 7-segment display and an internal multiplexing circuitry with four multiplexing outputs. The multiplexing circuit has its own free-running oscillator and does not require external clock. The counter advances on negative edge of the clock. A 'high' signal on display select (DS) input selects the number in the counter to be displayed; a low level signal on the DS selects the number in the output latch to be displayed.

A maximum of 9999 digits can be counted and shown on DIS1 through DIS4. A high signal at reset pin 13 of IC3 turns all the outputs to zero. So the display shows 0000 when

stationary part such that the magnet passes close to the reed switch at each rotation. Now press reset switch S2 momentarily. The display shows 0000. Switch on the motor and note the displayed reading after one minute to see the rpm.

Construction and testing

An actual-size, single-side PCB for the rotation counter is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable plastic case. After all the components are soldered on the PCB, connect the two wires coming from the reed switch to the main unit.

To check the circuit for proper functioning, verify 12V supply at TP1 and 5V at TP2, with respect to TP0. The clock pulses corresponding to the rotation can be verified at TP3. The reset to the circuit can be observed at TP4. Power LED1 should glow when connected to power supply. ●

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